An Elementary Existence Correction and Finite Minimum Formula for OEIS A115038 and Its Companions

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Abstract

The existence conjecture recorded in A115038, A115039 and A115041 follows from the elementary identity $((p-1)/2)^2+p=((p+1)/2)^2$ for odd primes p, together with $5^2+2=3^3$. The least solution can be improved on the square witness only by an odd prime power. This gives a finite exact formula for every term and proves that all three companion sequences are defined. The three entries encode the same existence question.

1 Existence and the unique square witness

For a prime p, let $X(p)$ be the least positive integer x for which $x^2+p=y^k$ with $y\geq 2$, $k\geq 2$. OEIS A115038 lists $X(p_n)$; A115039 gives the largest exponent k at that least x, and A115041 gives the corresponding base y [1]. All three entries currently repeat the conjecture that a solution exists for every prime.

Theorem 1. $X(2)=5$. *For every odd prime p,*

$$1\leq X(p)\leq (p-1)/2.$$

The only positive solution of $x^2+p=y^2$ is $x=(p-1)/2$, $y=(p+1)/2$.

Proof. For odd p, expand

$$\left(\frac{p-1}{2}\right)^2+p=\left(\frac{p+1}{2}\right)^2.$$

Conversely $(y-x)(y+x)=p$ has the unique positive factor pair $(1,p)$, which gives the asserted solution. For $p=2$, the values at $x=1,2,3,4$ are $3,6,11,18$, none a perfect power, while $5^2+2=27=3^3$. □

2 A finite minimum formula

For an odd prime p, put $U = (p-1)/2$ and $B = (p+1)^2/4$. For every odd prime $r \le \lfloor \log_2(B-1) \rfloor$, consider the finite integer interval

$$\left\lceil (p+1)^{1/r} \right\rceil \le y \le \left\lfloor (B-1)^{1/r} \right\rfloor. \tag{1}$$

Let $\mathcal{C}(p)$ consist of U together with the positive integers $\sqrt{y^r - p}$ for pairs (r, y) in these ranges for which $y^r - p$ is a perfect square. Integer roots are taken exactly.

Theorem 2 (Exact minimum)**.** *For every odd prime p,*

$$X(p) = \min \mathcal{C}(p). \tag{2}$$

Together with $X(2) = 5$, this determines every term of A115038 by finite root operations and finite minimization.

Proof. Every member of $\mathcal{C}(p)$ is an eligible positive x. If an eligible x is smaller than U, its value $V = x^2 + p$ satisfies $p < V < B$. Write $V = z^k$ with $k \ge 2$, and choose a prime divisor r of k. If $r = 2$ then V is square, contradicting the uniqueness assertion of Theorem 1. Hence r is odd, and $V = (z^{k/r})^r$ gives a pair in (1). The lower bound follows from $V \ge p + 1$, the upper bound from $V \le B - 1$, and $z^{k/r} \ge 2$ implies $r \le \log_2(B-1)$. Thus every possible improvement appears. □

For the companion sequences set $V = X(p)^2 + p$ and define

$$E(p) = \max\{k \in \{2, \dots, \lfloor \log_2 V \rfloor\} : \lfloor V^{1/k} \rfloor^k = V\}, \qquad Y(p) = \lfloor V^{1/E(p)} \rfloor.$$

These sets are finite and nonempty. Thus $\mathrm{A115039}(n) = E(p_n)$ and $\mathrm{A115041}(n) = Y(p_n)$ exist at every positive index. Equivalently, $E(p)$ is the gcd of the exponents in the prime factorization of V. For example $p = 31$ gives $X(p) = 1$, $V = 32$, $E(p) = 5$, $Y(p) = 2$.

The existence identity is a standard difference-of-squares observation; this note supplies a direct correction to the recorded conjecture and a finite least-value reduction. The supplied verifier compares the formula with an independent scan in x, using factorization to recognize perfect powers and their maximal exponents.

References

[1] C. Hilliard, OEIS A115038, A115039 and A115041, https://oeis.org/A115038, https://oeis.org/A115039, https://oeis.org/A115041, accessed October 26, 2026.